AtliQ Hardwares – SQL Challenge Insights

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Problem Statement

- Management lacked sufficient data insights for quick, smart decision-making
- Needed to expand analytics team with technically strong candidates
- A SQL challenge was designed to evaluate both technical and soft skills
- Goal: Answer ad hoc business requests using SQL to drive data-informed decisions

Tools & Technologies

Tools :

- MySQL — Database querying & analysis
- Canva— Presentation & visualization
- PowerBi/Charts — Data visualization support

SQL Concepts Applied :

- JOINs (INNER, LEFT)
- CTEs — WITH clause
- Window Functions — RANK, DENSE_RANK
- CASE Statements
- Subqueries
- Aggregate Functions — COUNT, AVG, SUM, MIN, MAX
- GROUP BY, HAVING, ORDER BY, LIMIT

Dataset Overview

Dimension Tables	Fact Tables	Key Details
dim_customer	fact_sales_monthly	FY: 2020 & 2021
dim_product	fact_manufacturing_cost	Focus: APAC & India
dim_market	fact_pre_invoice_deductions	Segments: 6
—	fact_gross_price	AtliQ Hardwares

Question 1:

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region

customer_code	market	customer	region
70002017	India	Atliq Exclusive	APAC
70003181	Indonesia	Atliq Exclusive	APAC
70004069	Japan	Atliq Exclusive	APAC
70006157	Philiphines	Atliq Exclusive	APAC
70007198	South Korea	Atliq Exclusive	APAC
70008169	Australia	Atliq Exclusive	APAC
70009133	Newzealand	Atliq Exclusive	APAC
70010047	Bangladesh	Atliq Exclusive	APAC
90002011	India	Atliq Exclusive	APAC

Insight:

Atliq Exclusive is an APAC customer, operating in the following markets:

- India
- Indonesia
- Japan
- Philiphines
- South Korea
- Australia
- Newzealand
- Bangladesh

This confirms its strategic presence in APAC, which makes it a high-value regional partner.

Question 2:

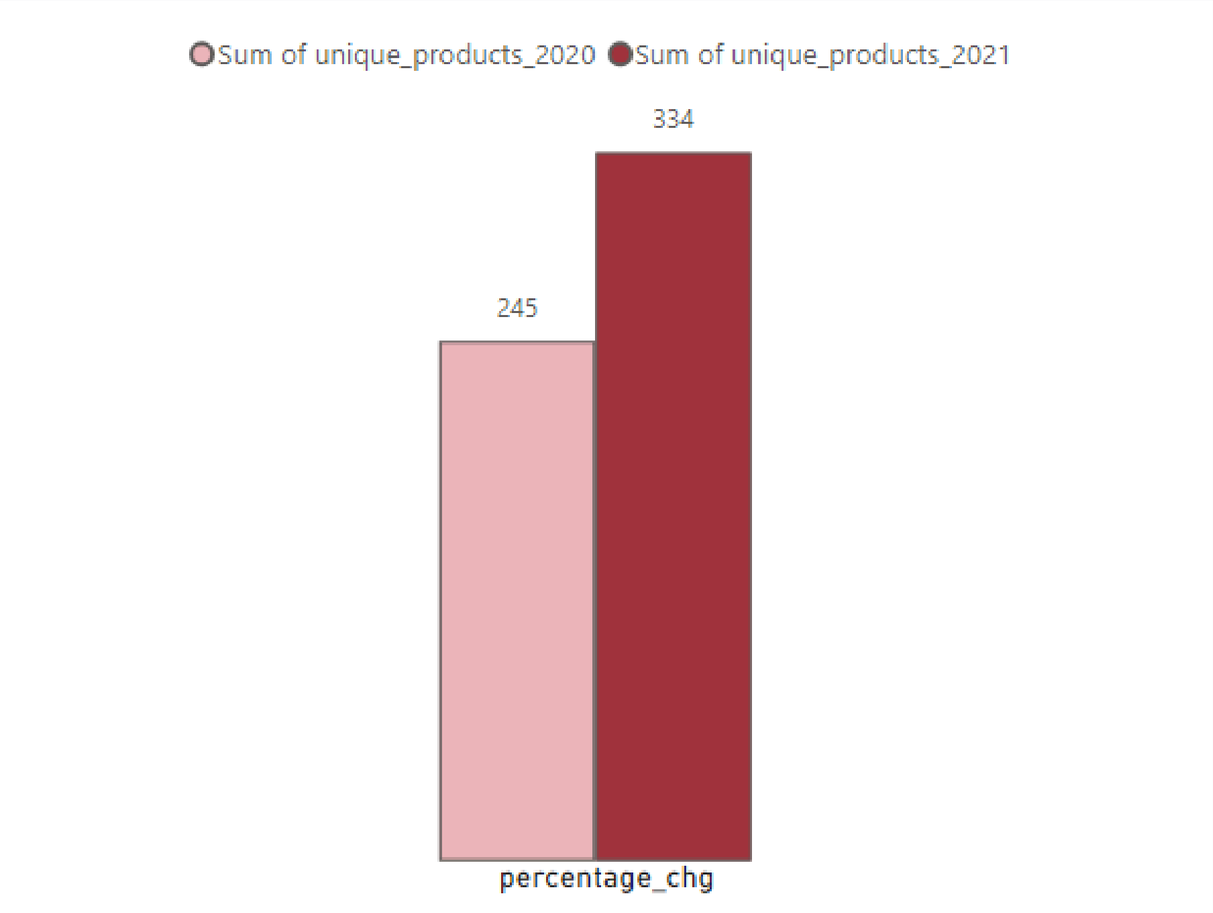
What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields, unique_products_2020 unique_products_2021 percentage_chg

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33%

In 2021, Atliq saw a Z% change in unique products compared to 2020. This highlights [growth/streamlining], which directly ties to the company's strategy in product innovation and market responsiveness.

Insight:

In 2021, Atliq saw a 36.33% change in unique products compared to 2020. This highlights growth, which directly ties to the company's strategy in product innovation and market responsiveness.



Question 3:

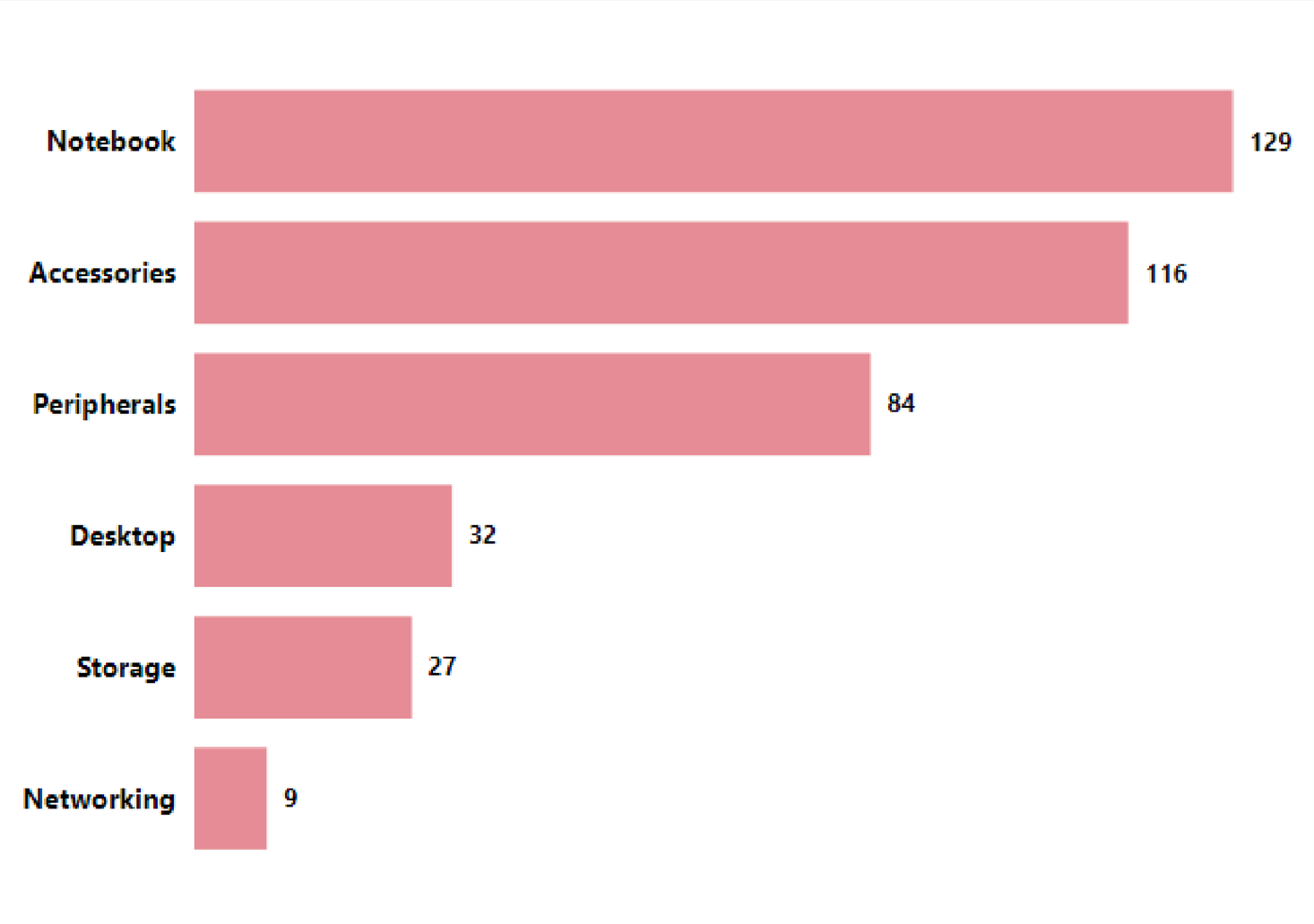
Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, segment product_count

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

Insights:

- The segment with the highest product_count (say Notebook with 120 unique products) is the broadest portfolio area, meaning the company focuses heavily on this category.

- Low product count → opportunity for expansion or deliberate streamlining to focus on core products



Question 4:

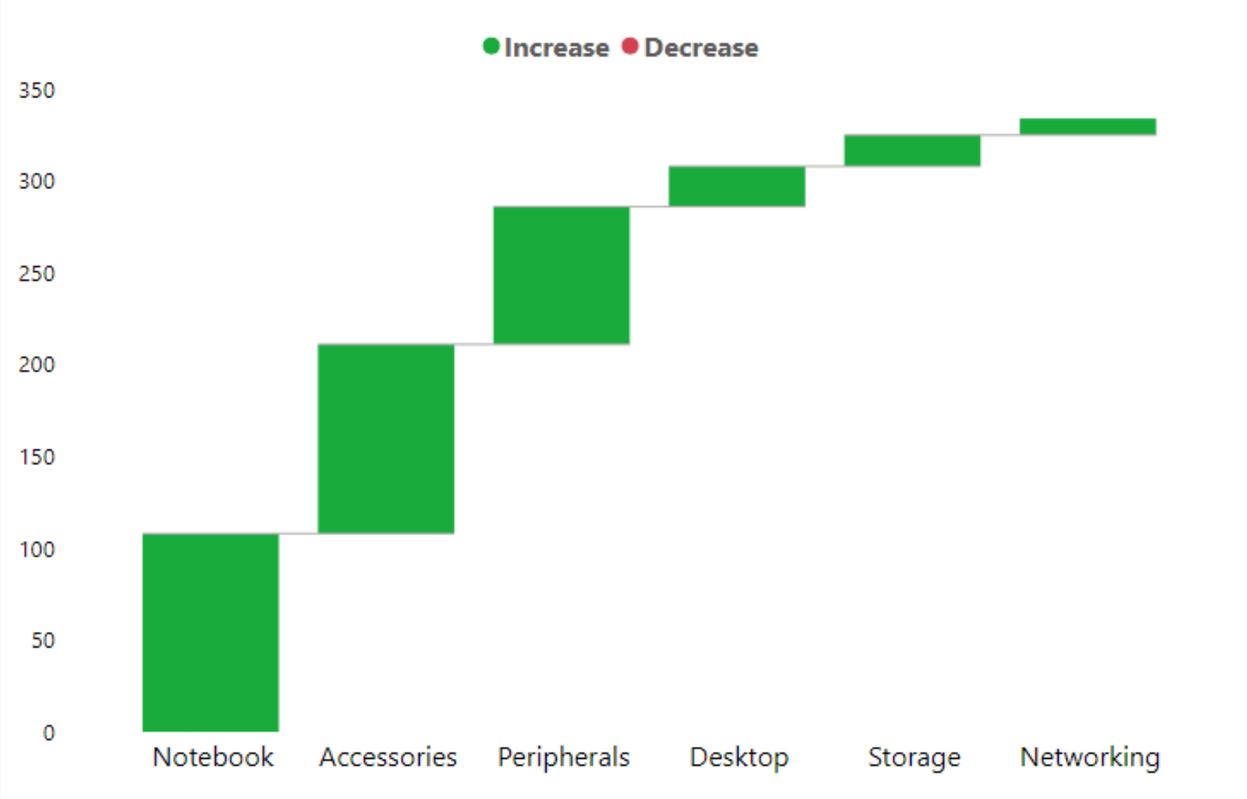
Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment product_count_2020 product_count_2021 difference

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Desktop	7	22	15
Networking	6	9	3
Notebook	92	108	16
Peripherals	59	75	16
Storage	12	17	5

Insights:

- In 2021, Accessories had the largest portfolio expansion, adding 34 new unique products compared to 2020. This highlights Atliq's strategy of strengthening its accessory ecosystem to complement core products.

- Networking and Storage grew steadily, but remain smaller focus areas.



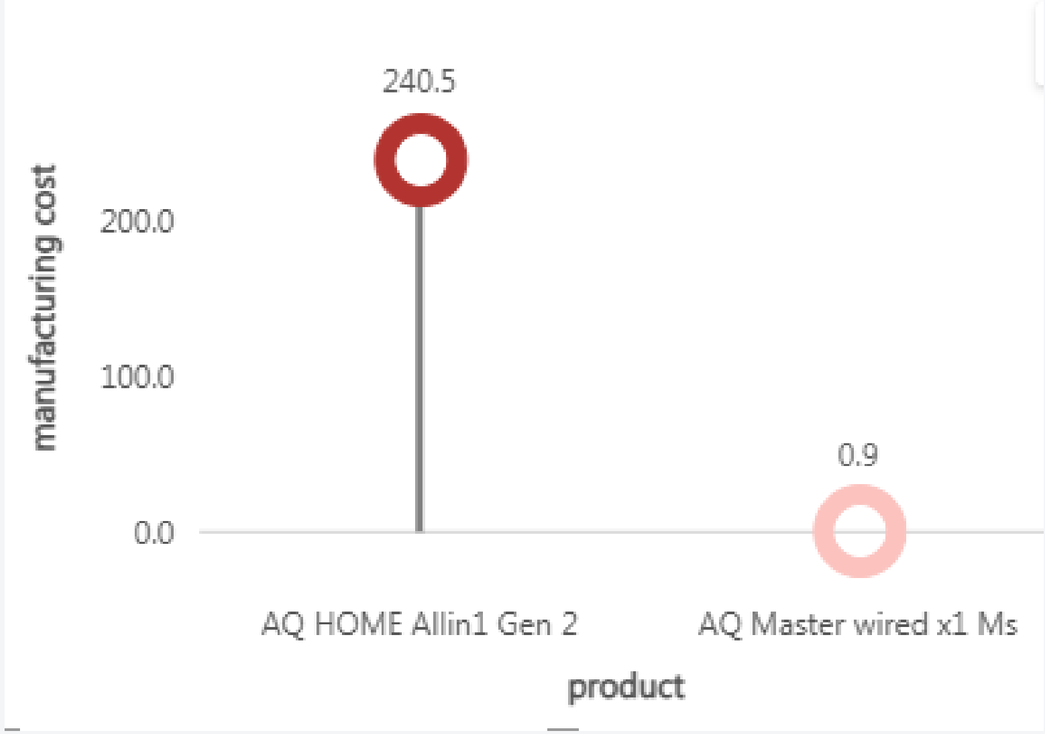
Question 5:

Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields, product_code product manufacturing_cost

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Insights:

The highest manufacturing cost product is **AQ HOME Allin1 Gen 2** at 240.54, while the lowest is **AQ Master wired x1 Ms at 0.89. This** showcases Atliq's wide product spectrum — balancing premium, high-complexity devices with affordable, mass-market accessories.



Question 6:

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields, customer_code customer average_discount_percentage

customer_code	customer	average_discount_percentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933

Insights:

- Flipkart received the highest discount (**30.83%**), indicating strong negotiation power and likely higher purchase volumes.
- The discount percentages are all close (**29–31%**), suggesting a consistent discount strategy across major Indian customers.

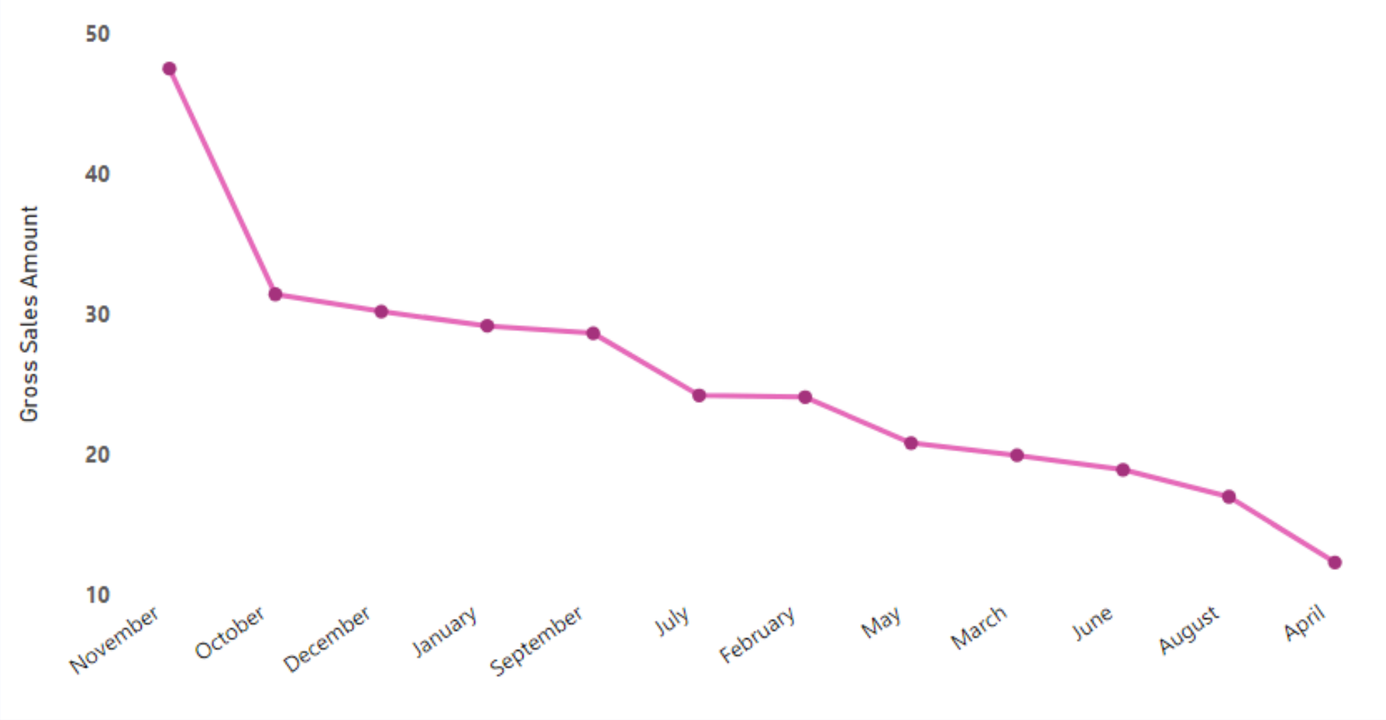
Question 7:

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: Month ,Year ,Gross sales Amount

Month	Year	Gross sales Amount
September	2019	9.09m
October	2019	10.38m
November	2019	15.23m
December	2019	9.76m
January	2020	9.58m
February	2020	8.08m
March	2020	0.77m
April	2020	0.80m
May	2020	1.59m

Insights:

- November shows the highest sales (~50), likely due to festive season demand (Diwali/holiday shopping).
- Strong sales are clustered around festival months (Oct–Nov), while summer months (Apr–Jun) show the lowest performance.



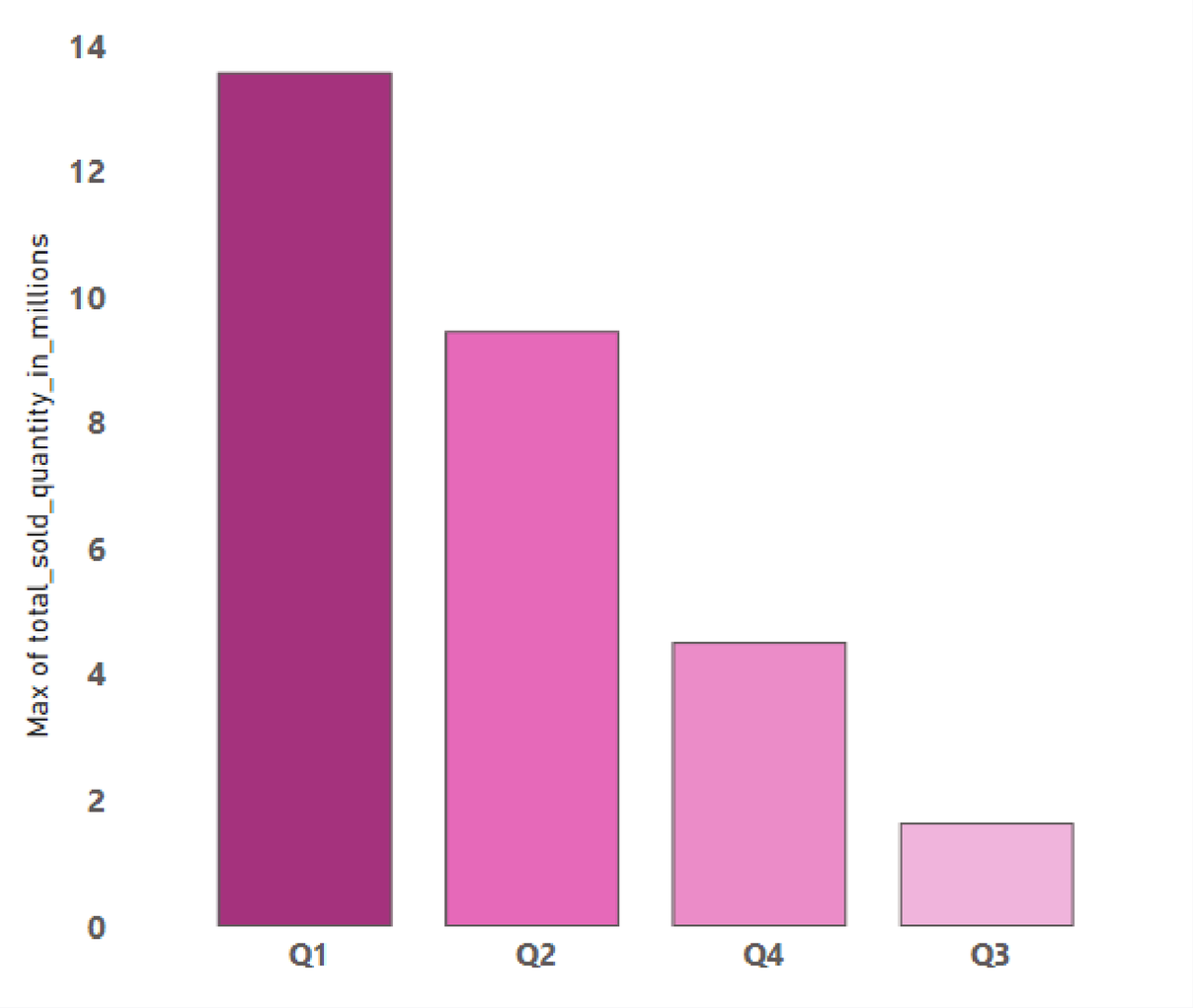
Question 8:

In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity, Quarter total_sold_quantity

fiscal_quarter	total_sold_quantity_in_millions
Q1	13.58
Q2	9.47
Q4	4.52
Q3	1.65

Insights:

- Q1 (13.58M) had the highest sold quantity, indicating very strong demand early in the year
- Q2–Q3 decline → May reflect COVID-19 impact in 2020.
- Q4 recovery → Indicates some market stabilization post-lockdown.



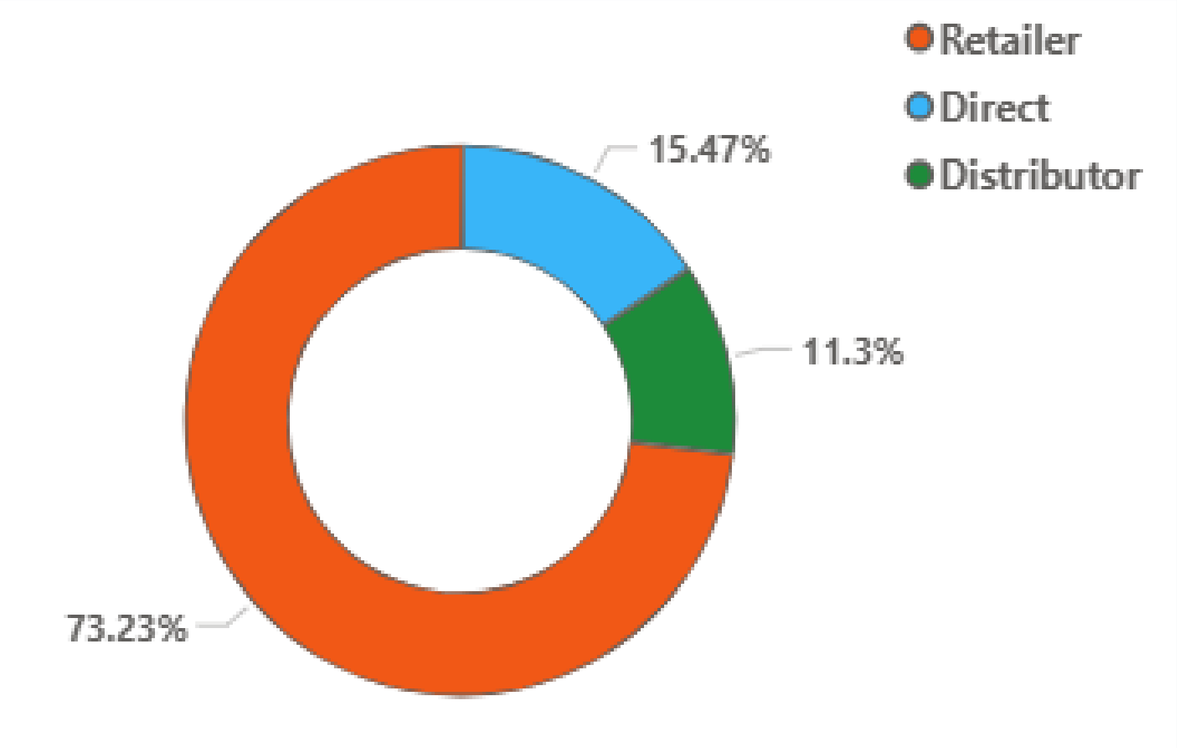
Question 9:

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel gross_sales_mln percentage

channel	gross_sales_mln	percentage_contribution
Retailer	1219.08	73.23%
Direct	257.53	15.47%
Distributor	188.03	11.30%

Insights:

- Retailer is the dominant channel with **73.23%** contribution — generating ₹1219.08M in gross sales
- Direct channel contributes **15.47%** — a secondary but significant revenue stream
- Distributor contributes the least at **11.30%** — smallest but still relevant channel



Question 10:

0. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields, division, product_code , product total_sold_quantity ,rank_order

division	product_code	product_name	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

Insights:

- Division N & S leads in volume with Pen Drive products showing strong demandin this region.
- Division PC has significantly lower total sold quantities (around 17,000 units) perhaps a smaller or more specialized market segment.

product_name	N & S	P & A	PC
AQ Digit			34,709.00
AQ Gamers Ms		4,28,498.00	
AQ Maxima Ms		8,39,336.00	
AQ Pen Drive 2 IN 1	7,01,373.00		
AQ Pen Drive DRC	13,64,248.00		
AQ Velocity			17,280.00
Total	**20,65,621.00**	**12,67,834.00**	**51,989.00**

Key Business Insights:

1. Atliq Exclusive operates in 8 APAC markets
2. Product portfolio grew **36.33%** — Accessories led expansion
3. November = Peak sales | Apr–Jun = Lowest performance
4. Retailer dominates with **73.23%** of FY2021 gross sales
5. Q1 2020 strongest quarter — Q2–Q3 impacted by COVID-19
6. Cost range ₹0.89 to ₹240.54 — wide product spectrum

Strategic Recommendations:

1. Invest in Accessories & Notebook — highest growth segments
2. Launch festive campaigns from September to capture November peak
3. Strengthen Retailer partnerships — primary revenue driver
4. Optimize discount strategy in Indian market
5. Targeted promotions during Summer to reduce revenue dip

THANK YOU !

"DATA TELLS A STORY — THE GOAL IS TO MAKE IT HEARD"